

Shared evidence | Light investigation

Photosynthesis: the light lab

Name _____ Date _____

Constructed sample data. Same shoot, solution, one-minute intervals and 20°C water. Sensor readings are mean lux at the vessel; lux is a practical light indicator.

Distance(cm)	Sensor(lux)	Three bubble counts in 1 min
40	300	4,5,6
30	600	9,10,11
20	1200	16,17,18

Method: use prepared shoot and same solution. Set light position; allow at least 2 minutes to adjust. Monitor temperature. Take three equal one-minute counts. For logging: sensor faces lamp in the same orientation; record once each second for 30 seconds and take its mean. Repeat at each position.

Paper route: analyse these values only. Optional demonstration uses teacher-prepared apparatus; record any real observations separately. Bubble sizes vary and some oxygen remains dissolved.

W1. Predict and calculate means. W2. Repair the two-variable method. W3. State a supported conclusion.

Supported route | Build the brief

Word bank: carbon dioxide; water; glucose; oxygen; light; same; rate; proxy.

S1. Write the photosynthesis word equation. Put the energy source above the arrow.

S2. Calculate each mean bubble count. Show one addition and division.

S3. Name what changes and two things kept the same. Why keep them the same?

S4. Describe the pattern with two values. State one reason counts are not exact oxygen volumes.

Standard route | Investigation proposal

Use the constructed table and precise units.

M1. State the photosynthesis word equation and explain the role of light/chlorophyll.

M2. Calculate all means and compare the lowest/highest light levels numerically.

M3. Describe a reproducible method: intended variable, measured outcome, timing/repeats, controls and sensor use.

M4. Write a supported conclusion and one improvement to the oxygen measurement. Explain why it helps.

Stretch route | How strong is the evidence?

Optional extension. A second constructed trial at 1200 lux gives one-minute counts 16,17,35. A note says the third bubble count may include two minutes of observation.

T1. Calculate the mean of all three counts. Explain why the timing note matters before accepting it.

T2. Describe a fair way to investigate the suspicious result without deleting it just because it is different.

T3. A light sensor rises further but the mean count stops rising. Suggest a possible limiting factor and a test that changes only that factor.

T4. Explain why lux and bubble counts are useful indicators but neither is a direct measurement of glucose production.

Exit and reflection | Process and measurement

Use one accurate sentence for each distinction.

E1. Name the two material reactants in photosynthesis.

E2. 15 bubbles in 3 minutes: show the rate calculation and unit.

E3. Give one control variable and one limitation of the bubble proxy.

L1. What did your audience ask? Improve one claim or measurement label.

Next useful step

- A. Separate matter from energy.
- B. Calculate a rate or mean.
- C. Control an unwanted variable.
- D. Explain the proxy limit.